

Tip Calculator PWA — System Documentation

A simple PWA tip calculator with bill splitting, multiple tip rates, receipt scanning, and share functionality. Replaces the native Tip Calculator app (no 7-day TestFlight expiry).

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Table of Contents

1. Document Control
2. System Overview
3. System Identity
4. Purpose and Intended Use
5. Builder / Build History
6. Hardware Prerequisites
7. Software Prerequisites
8. Capabilities
9. Limitations
10. Architecture Overview
11. Dependencies
12. File and Folder Locations
13. Core Workflows
14. End-User Operating Guide
15. Option 1 — Operate via Claude
16. Option 2 — Operate Manually via CLI
17. Option 3 — Operate via PWA (Primary)
18. Outputs

- 19. Housekeeping and Maintenance
- 20. Troubleshooting
- 21. Off-Peak / Remote / Scheduled Operation
- 22. Change Log
- 23. Appendix

1. Document Control

Field	Value
Document title	Tip Calculator PWA — System Documentation
Version	1.0
Status	Active
Owner	Doug
Last updated	2026-06-17
Storage location	`~/Desktop/Claude Summary/Tip Calculator PWA/docs/SYSTEM_DOCUMENTATION.md`
Related files	`docs/README.md`

2. System Overview

Summary

A lightweight Progressive Web App that calculates tip amounts for restaurant bills. Supports multiple tip percentages (12%, 15%, 18%, 20%), bill splitting across multiple people, receipt photo scanning (OCR), and sharing results.

Executive snapshot

Item	Summary
System name	Tip Calculator PWA
Main purpose	Calculate tips and split bills
Primary user(s)	Doug (personal use)
Primary interface(s)	PWA on iPhone (Add to Home Screen)
Current status	Active — replaces native Tip Calculator

3. System Identity

Field	Details
System name	Tip Calculator PWA
Short name	TipCalc
Project folder	~/Desktop/Claude Summary/Tip Calculator PWA/
Runtime environment	Any modern browser (Safari, Chrome)
Main language(s)	HTML, CSS, JavaScript
Primary execution mode	PWA served via local HTTP server

4. Purpose and Intended Use

Purpose

Quickly calculate tip amounts and per-person shares when dining out.

Intended use cases

- Calculate tip at multiple rates (12%, 15%, 18%, 20%) for a bill
- Split the total (bill + tip) across multiple people
- Scan a receipt photo to auto-fill the bill amount
- Share the tip breakdown with dining companions

Out-of-scope / non-goals

- Tax calculation
- Multi-currency support
- Bill itemization or per-person ordering

5. Builder / Build History

Question	Details
Who built the system?	Doug + Claude Code
When was it built?	March 2026
How was it built?	Claude Code Opus 4.6
What was the build approach?	PWA to replace native iOS app (avoids TestFlight 7-day expiry)
Subsequent modifications	Folder restructure to standard layout (2026-06-16)

6. Hardware Prerequisites

Component	Minimum requirement	Recommended	Notes
Mac	Any Mac	Any Mac	To serve PWA locally
iPhone	Any with Safari	iPhone with camera	For receipt scanning

7. Software Prerequisites

Component	Required version	Purpose	Notes
Python 3	3.x	Local HTTP server	Built-in on macOS
Browser	Safari / Chrome	Run the PWA	Any modern browser

Installation notes

No installation needed beyond serving the files. For iPhone home-screen install:

1. Serve the pwa/ folder via HTTP
2. Open the URL in iPhone Safari
3. Share → Add to Home Screen

8. Capabilities

Capability	Description	User-facing?	Notes
Tip calculation	Compute tip at 12%, 15%, 18%, 20%	Yes	All rates shown simultaneously
Bill splitting	Divide total by number of people (1–99)	Yes	+/- buttons
Receipt scanning	OCR via camera to extract bill amount	Yes	Camera button
Share results	Share tip breakdown via OS share sheet	Yes	Share button

9. Limitations

Limitation	Impact	Workaround	Notes
Fixed tip rates	Only 12/15/18/20%	N/A	Covers common rates
USD only	No currency conversion	N/A	Personal use

Limitation	Impact	Workaround	Notes
Requires server	Must serve files via HTTP for PWA features	Run <code>`python3 -m http.server`</code>	Service worker needs HTTP
OCR accuracy	Receipt scanning may misread amounts	Manual entry	Depends on photo quality

10. Architecture Overview

High-level architecture

Single-page PWA with no backend. All logic runs client-side in the browser.

Architecture table

Layer / component	Function	Inputs	Outputs	Notes
index.html	Page structure	N/A	DOM	Entry point
style.css	Visual styling	N/A	Styled UI	iOS-native look and feel
app.js	Business logic	Bill amount, split count	Tip grid	All calculation logic
sw.js	Service worker	Fetch events	Cached responses	Offline support
manifest.json	PWA manifest	N/A	App metadata	Home screen install

Data flow

1. User enters bill amount (manual or OCR scan)
2. `app.js` computes tip for all 4 rates, divided by split count
3. Results displayed in grid (per-person or total)
4. User can share results via OS share sheet

11. Dependencies

Dependency	Version	Why needed	Where used
Python 3	3.x	Local HTTP server	Serving PWA files

No external libraries — pure HTML/CSS/JS.

12. File and Folder Locations

Path	Purpose	Editable by user?	Notes
`pwa/index.html`	Main app page	No	HTML structure
`pwa/style.css`	Styles	No	iOS-native appearance
`pwa/app.js`	App logic	No	Tip calculation, OCR, sharing
`pwa/sw.js`	Service worker	No	Offline caching
`pwa/manifest.json`	PWA manifest	No	App name, icons, theme
`pwa/icon.png`	App icon	No	Home screen icon
`docs/`	Documentation	No	This file, README

13. Core Workflows

Workflow	Objective	Main steps	Output
Calculate tip	Get tip amounts	Enter bill → view grid	Tip at 4 rates
Split bill	Per-person amount	Enter bill → adjust people count	Per-person totals
Scan receipt	Auto-fill bill amount	Tap camera → take photo	Bill amount populated
Share results	Send to others	Tap share button	Shared via OS share sheet

14. End-User Operating Guide

Overview

Open the Tip Calculator from your iPhone home screen (or browser). Enter the bill amount, adjust the number of people if splitting, and read the tip amounts from the grid.

Quick start

1. Enter the bill amount in the \$ field
2. The grid instantly shows tip and total at 12%, 15%, 18%, and 20%
3. Use +/- to adjust the number of people splitting the bill
4. Tap the camera icon to scan a receipt instead of typing
5. Tap share to send the breakdown

15. Option 1 — Operate via Claude

N/A — this is a simple calculator app. Claude can help modify the code but doesn't operate the app.

Useful prompts

```
Add a custom tip percentage option to the Tip Calculator PWA. The project is at  
~/Desktop/Claude Summary/Tip Calculator PWA/
```

16. Option 2 — Operate Manually via CLI

Start the server

Primary method (recommended):

```
bash ~/Desktop/Claude\ Summary\ Tip\ Calculator\ PWA/start.sh
```

Manual alternative:

```
cd ~/Desktop/Claude\ Summary\ Tip\ Calculator\ PWA  
python3 pwa/serve.py
```

Access on iPhone

Open `http://<mac-ip>:8093` in Safari → Share → Add to Home Screen.

17. Option 3 — Operate via PWA (Primary)

User steps

1. Tap the Tip Calculator icon on your iPhone home screen
2. Enter the bill amount
3. Adjust split count if needed
4. Read tip amounts from the grid
5. Share if desired

This is the primary and recommended way to use the app.

18. Outputs

N/A — the app displays results on-screen. No files or reports are generated.

19. Housekeeping and Maintenance

Task	Frequency	How to do it	Why it matters
Clear browser cache	Rarely	Safari Settings → Clear cache	If PWA behaves oddly
Re-add to home screen	After server IP change	Share → Add to Home Screen	URL may change

Minimal maintenance required — this is a simple, self-contained PWA.

20. Troubleshooting

Problem	Likely cause	How to diagnose	How to fix
App won't load	Server not running	Check terminal	<code>`bash start.sh`</code>
Camera not working	No HTTPS (required for camera API)	Check URL	Use localhost or set up HTTPS
Stale version after update	Service worker cache	Check sw.js version	Increment cache version in sw.js
Can't add to home screen	Not served via HTTP	Check URL bar	Must use http:// not file://

21. Off-Peak / Remote / Scheduled Operation

N/A — this is an on-demand calculator app. No scheduled or background operation.

22. Change Log

Date	Author	Change summary
2026-03-28	Doug + Claude	v1.0 — Initial PWA release
2026-06-16	Claude	Folder restructure to standard layout
2026-06-17	Claude	Created SYSTEM_DOCUMENTATION.md

23. Appendix

Appendix A — Tip rates

Default rates: 12%, 15%, 18%, 20%. These are defined in `app.js` as the `tipRates` array.

Appendix B — PWA install

For the best experience, add to iPhone home screen via Safari's Share → Add to Home Screen. This gives the app a native app-like appearance with no browser chrome.